AUTHBC: Co-Optimizing Encoding, Authentication Placement, Signature Scheme, and Batching for Blockchain-Grade UAV Telemetry over 802.11

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AUTHBC Project — Thesis draft (auto-generated from frozen results)

***Abstract*—Flying ad-hoc networks (FANETs) stream tiny telemetry records that must be individually authenticated and hash-chained into a tamper-evident ledger. Because a cryptographic signature (64–96 B) rivals or exceeds the payload (45–190 B), naive per-record signing wastes airtime, energy, and goodput on a shared, lossy 802.11 channel. We argue that four design knobs—record *encoding*, authentication *placement*, signature *scheme*, and *batch* size—are coupled and must be *co-optimized*. We formalize this with five theorems (overhead dominance, batching amplification and its regime boundary, a loss-robustness frontier, a power-independent scheme crossover, and the co-design optimum), implement measurable models for each, and test one falsifiable criterion: the co-optimized configuration must cut on-air authentication bytes by $\geq 40\%$ versus an inline-signature CBOR baseline while keeping verifiability $V \geq 0.95$ under per-frame loss $p = 0.05$. Driven by measured microbenchmarks, the optimizer selects a delta encoding under self-batch placement, cutting authentication overhead by 75.0% (to 26.0 B/record) at $V = 0.95$ *and* a 250 ms freshness bound—criterion met. Freshness, not the MTU, is what binds: the byte-optimal batch cuts 96.8% but incurs 1.55 s of staleness, so a co-design optimizer blind to freshness solves the wrong problem. The airtime model is validated against NS-3 3.41: unicast matches Bianchi within $+0.6/-2.9\%$, while the common reduction of that model to broadcast fails by $16\times$ at $N{=}50$ and is replaced by Ma and Chen's broadcast model [1], [2], which our measurements confirm to within 0.36%. All results are reproduced deterministically from frozen data by an automated staleness gate. AUTHBC introduces no new primitive; its contribution is the rigorous, reproducible co-design and its quantified trade-offs.**

***Index Terms*—FANET, UAV telemetry, authentication, signature aggregation, blockchain, CBOR, 802.11, Bianchi model.**

I. Introduction

A fleet of UAVs streams telemetry—position, velocity, altitude, battery, mode—at $\Lambda \approx 20$ records/s per node over 802.11 into a blockchain-grade, hash-chained, signed ledger. Each record must be *authenticated* (a signature so a verifier trusts its origin) and *chained* (a hash of the previous record, so the log is tamper-evident).

The difficulty is scale mismatch. Telemetry records are tens of bytes; an Ed25519/ECDSA signature is 64 B, a SHA-256 chain hash is 32 B, and a BLS signature is 96 B. Signing every record inline makes authentication overhead rival or exceed the payload, and on a shared, contended, lossy channel that wasted airtime directly costs goodput, energy, and freshness. How much overhead one actually pays depends on four *coupled* choices: the record *encoding* e (setting the size s), the authentication *placement* (inline, self-batch, relay-aggregate, or block-level), the signature *scheme* σ, and the *batch* size b. Our thesis is that these knobs are not separable and that co-optimizing them jointly beats tuning any one. We make this precise (Sec. III), implement it (Sec. IV), and test the falsifiable headline of Sec. V.

II. Related Work and Positioning

Citation integrity. Foundational works below are cited with verified detail; entries marked **[VERIFY]** denote domain-specific literature whose exact references must be confirmed against primary sources—no figure is quoted from an unverified source.

(a) Signatures for constrained authentication. ECDSA [3] yields 64 B P-256 signatures at 128-bit security but is randomness-sensitive; Ed25519 [4], [5] uses a twisted-Edwards curve with deterministic nonces for fast, misuse-resistant, batchable verification; BLS [6] gives the shortest signatures via pairings, and [7] aggregates n signatures into one at the cost of expensive pairing verification. AUTHBC introduces *no new primitive*; it quantifies *where and when* each scheme wins once placement and batching are fixed (Sec. III-D), finding Ed25519 dominant for self-batch and BLS worthwhile only for cross-signer relay traffic.

(b) Batch/aggregate verification in vehicular/IoT networks [8] [VERIFY]. These amortize receiver-side verification across many messages. AUTHBC generalizes from receiver batching to authentication *placement* and co-designs it with encoding and scheme, and adds the loss-robustness frontier that pure aggregation ignores.

(c) Blockchain/hash-chained ledgers for UAV/IoT provenance [9] [VERIFY]. Nakamoto's hash chain [10] makes a log tamper-evident; UAV-blockchain systems place telemetry or attestations on such a chain, gaining integrity at per-record overhead cost. AUTHBC targets exactly that substrate byte/energy cost and is agnostic to the consensus layer.

(d) Compact serialization [11] [VERIFY for delta]. CBOR [12] and MessagePack [13] give compact binary encodings; differential/delta coding sends small deltas against periodic keyframes. AUTHBC measures these head-to-head and shows encoding is coupled to authentication: a smaller payload raises the auth fraction, which *increases* the value of batching.

(e) Analytical 802.11 modelling. Bianchi [14] models the DCF as a per-station Markov chain whose fixed point yields closed-form saturation throughput. AUTHBC uses it as the airtime cost model inside the optimizer and validates it against NS-3 [15].

Overall. AUTHBC is a rigorous, hardware-validated co-design and measurement study, not a new construction. Prior work typically fixes three of the four knobs; we jointly optimize all four under a single falsifiable criterion with a reproducibility gate.

III. System Model and Theory

A record is integer-only fixed-point (floats break canonical-CBOR determinism) with a 32 B `prev_hash` chain link; its encoded size is s. A frame carries b records, a header $H_f \approx 40$ B, and an authentication object g_a (64 B Ed25519/ECDSA, 96 B BLS), bounded by the MTU $M = 1500$ B. On-air bytes per record, by placement:

$$\text{A (inline)} : s + g_a + H_f/b, \tag{1}$$

$$\text{B (self-batch)} : s + (g_a + H_f)/b, \tag{2}$$

$$\text{D (block, } n \text{ frames)} : s + (g_a + nH_f)/b. \tag{3}$$

A. T1: overhead dominance

For inline authentication the pure-authentication fraction is

$$\varphi = \frac{g}{s+g}. \tag{4}$$

For $s = 45$–190 B and $g = 64$ B, $\varphi = 25$–59%: the smaller the payload, the more the signature dominates. This is the disease; T2–T5 are the cure and its limits.

B. T2: batching cure and amplification

Self-batch folds b records under one signature, so the per-record auth cost $(g_a + H_f)/b \to 0$. Packing to the MTU gives a goodput amplification over single-record framing:

$$A = \frac{M}{M - H_f - g_a}, \tag{5}$$

since at the MTU-filling batch $b_{\max}$ the frame $\approx M$ and $b_{\max} s \approx M - H_f - g_a$. Inline (A) can *never* amortize its per-record signature (only H_f/b shrinks).

T2a: when does A apply? The derivation assumes the *MTU* caps the batch. Under a freshness bound the cap is instead $b = \lfloor \Lambda D_{\max} \rfloor$, which does not depend on s; then $C(s) = s + (g_a + H_f)/b$ and $dC/ds = 1$ *exactly*—compression pays $1\times$, not $A\times$, and the residual authentication cost is a floor compression cannot reach. The boundary is $s < (M - H_f - g_a)/(\lfloor \Lambda D_{\max} \rfloor + 1)$. At Λ=20 rec/s and $D_{\max}$=250 ms this is 232.7 B on 802.11, above every encoding studied, so A **is never operative there**; on a low-rate link (M=222) the MTU binds again and A=1.88 *is* operative. The "compression pays $\times A$" leverage is therefore real and *exclusive to low-rate links*.

C. T3: loss-robustness frontier

Let p be per-frame loss and $V = \Pr[\text{a received record is independently verifiable}]$. Self-batch frames self-verify, so $V_B = 1 - p$, *independent of* b. A block signature spanning $n(b) = \lceil (bs + g_a + H_f)/M \rceil$ frames needs all fragments, so $V_D = (1-p)^n$. Whenever $n > 1$, $V_B = 1 - p > V_D = (1-p)^n$: self-batch Pareto-dominates the marginal byte saving of over-aggregation.

D. T4: power-independent scheme crossover

Per-record energy is $E = P_c\,(\text{CPU time}) + P_r\,(\text{radio time})$. Comparing Ed25519 and BLS, which wins depends on the powers *only* through their ratio $\kappa = P_r/P_c$:

$$\text{BLS wins} \iff \kappa > \kappa^* = \frac{\Delta\text{CPU}}{\Delta\text{RADIO}}, \tag{6}$$

where ΔCPU is BLS's extra CPU and ΔRADIO its radio-time saving. κ^* is the break-even power ratio, obtained without measuring a single watt. Wi-Fi receive power sits below CPU-active power, so plausible platforms have $\kappa \lesssim 0.5$.

E. T5: co-design optimum

Enumerate $(e, \sigma, \text{placement}, b)$ and take the Pareto front over $\{\text{min bytes, min energy, max } V\}$ subject to MTU fit, $V \geq 1-\epsilon$, and verify-throughput $t_{\text{verify}}(b)\,\Lambda \leq 1$. The optimum is the smallest deterministic encoding under self-batch at the largest MTU-feasible batch.

F. Channel and energy models

At the 6 Mb/s OFDM base rate, broadcast airtime is $T_{\text{air}}(L) = T_{\text{phy}} + 8(L + \text{MAC})/R + \text{DIFS} + \delta$ (fixed part $\approx 100\,\mu$s); unicast adds SIFS and an ACK ($\approx 156\,\mu$s). Under contention among N saturated stations, the Bianchi DCF fixed point is

$$\tau = \frac{2(1-2p_c)}{(1-2p_c)(W+1) + p_c W(1-(2p_c)^m)}, \quad p_c = 1-(1-\tau)^{N-1}, \tag{7}$$

with $W = 16$, $m = 6$, solved by damped iteration; saturation throughput $S = P_{tr} P_s\, 8L/E[\text{slot}]$. Self-batch per-record energy is $E = P_c(t_{\text{enc}} + t_{\text{sign}}/b + t_{\text{verify}}/b) + P_r\, T_{\text{air}}(\text{frame})/b$.

IV. Implementation

The project ran in phases under an eight-law discipline (plan-first, verify-before-assume, never-bypass-a-failure, TDD, audit–attack–fix, check-results, scientific-integrity, escalate-decisions); `main` is always green and every result carries an environment and config-hash header. **P0** locked the toolchain and CI. **P1** built four encoders and three schemes under a strict timing harness (GC off, warmup, checksum, bootstrap CIs) validated against real KATs (RFC 8032, Wycheproof, Chia). **P1b** switched the binary codecs to canonical schema arrays (CBOR 110→66 B). **P2** built the hash-chained store and frozen canonical-CBOR wire vectors. **P3** implemented the four placements and the 802.11a airtime model (broadcast and unicast kept separate). **P4** turned T1–T3 into seeded experiments (E1–E3). **P5** built the energy model, the exhaustive

TABLE I
E1: OVERHEAD DOMINANCE (T1), 30 SEEDS. $\varphi = 64/(s + 64)$.

encoding	mean bytes s	φ
JSON	191.1	25.1%
CBOR	66.25	49.1%
MessagePack	65.16	49.6%
delta	**45.00**	**58.7%**

Pareto optimizer, and the power-independent crossover (E4). **P6** validated the Bianchi model against NS-3 3.41. **E5** fed the optimizer measured inputs and tested the headline. **P7a** produced the hardware-measurement scripts; a whole-repo audit then corrected one inconsistency and added an automated frozen-data reproduction gate (Sec. VI).

V. RESULTS

All values are from frozen CSVs (x86 host, Intel i5-14400F; energy uses nominal power pending hardware measurement).

E1 (Table I). The auth fraction climbs from 25% (JSON) to 59% (delta): compressing the payload makes a fixed 64 B signature dominate—T1 confirmed.

P1 crypto (x86 medians). ECDSA-P256 sign/verify $26.2/78.5\,\mu$s; Ed25519 $29.1/95.0\,\mu$s; BLS $324.6/1016.5\,\mu$s; BLS aggregate-verify(b=8) $3445\,\mu$s. Notably, ECDSA *beats* Ed25519 on x86 (OpenSSL P-256 assembly)—the reverse of the usual ordering, and the reason the optimizer breaks a byte-tie toward ECDSA on the energy metric; this edge is expected to vanish on ARM.

E2 (T2/T2a). At MTU 1500, self-batch, the measured amplification $A_{@b}$ = (bytes/rec)/s matches $A = M/(M - H_f - g_a)$ within 5% (integer-b slack, shrinking with s); delta reaches $b_{\max} = 31$ with $A_{@b} = A = 1.0745$ exactly. The MTU sweep also straddles the T2a boundary: M=256 is MTU-limited (A=1.68 realised), while M=576 and M=1500 are freshness-limited and realise A=1.0000—i.e. the $b_{\max}$=31 above is *unreachable* at telemetry rates.

E3 (T3). $V_B \approx 1 - p$ flat in b (0.90 at p=0.10); $V_D = (1-p)^n$ drops the moment a block spans $n > 1$ frames (0.81 at p=0.10, b=40)—the frontier is exactly T3.

E4 (T4, Fig. 1). Ed25519 verify is 10.7× cheaper than BLS verify. Across the full relay-fraction × batch × rate grid (80 points), Ed25519 wins *every* one (minimum $\kappa^* = 3.21 \gg 0.5$). At 96 B, BLS carries more auth bytes than Ed25519 on own self-batch traffic ($\kappa^* = \infty$) and saves bytes only on relay traffic for $b \geq 2$; its 10.7× verify cost keeps Ed25519 optimal for any plausible power ratio.

E5 (Table II, Fig. 2). The optimizer selects delta + Ed25519 + self-batch at b=4, cutting on-air authentication from 104 B/record to **26.0 B/record**—a **75.0%** reduction versus the Pillar-1 baseline at $V = 0.95$, $p = 0.05$ and a 250 ms freshness bound. Hand-checked: $45.0 + 104/4 = 71.0$ B/record, freshness $4/20 = 200$ ms.

Which constraint binds is itself the result. Ignoring freshness, the byte-optimal batch is b=31 at the MTU knee, cutting 96.8%—but the oldest record in such a batch waits

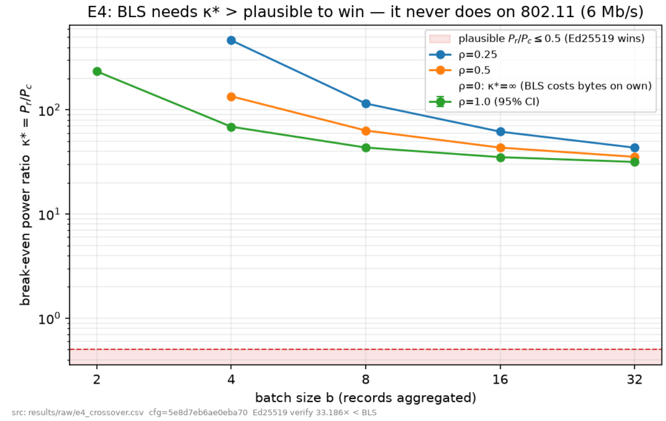

Fig. 1. E4: break-even power ratio $\kappa^*(b)$ per relay fraction. Every curve sits far above the plausible band ($\kappa \leq 0.5$), so Ed25519 wins on 802.11; the ρ=0 curve is ∞ (BLS costs bytes on own traffic).

TABLE II
E5: CO-DESIGN HEADLINE (T5) UNDER $V \geq 0.95$, p=0.05 AND $D \leq 250$ MS. AUTH CUT $= (104 - 26.0)/104 = 75.0\% \geq 40\% \Rightarrow$ PASS.

config	enc	σ	place	auth B/rec	V	D (ms)
optimized	delta	Ed	B,b=4	**26.00**	0.95	200
A+CBOR (Pillar-1)	cbor	Ed	A	104.0	0.95	50
A+JSON (naive)	json	Ed	A	104.0	0.95	51
D-over-agg	cbor	Ed	D,b=40	3.60	0.9025	2004

1.55 s, 6.2× over $D_{\max}$. Because fill time dominates the delay, the admissible batch obeys $b \lesssim \Lambda D_{\max}$ *independent of encoding and scheme*, so freshness binds long before the MTU does at telemetry rates. The co-design answer is therefore a frontier, not a point: b=1→4 trades $0/50/66.7/75\%$ auth-byte reduction against $50/100/150/200$ ms of staleness, with sharply diminishing returns. Over-aggregation (D) is byte-competitive (3.60 B) but *fails* $V \geq 0.95$ ($0.9025 = (1-p)^2$)—a live demonstration of the T3 frontier—and misses freshness by 8×.

NS-3 validation (Table III, Fig. 3). Unicast validates the Bianchi model [14] to $+0.6/{-2.9}\%$ across N = 5–50. Broadcast requires a *different* model, not the same one with the ACK removed. Reducing the unicast model by fixing $\tau = 2/(W{+}1)$ under-predicts NS-3 by 16× at N=50. PHY-trace instrumentation shows the cause is not frame capture—which we measure at **0%**, every successful decode arriving from a single-transmitter busy period—but the *backoff counter consecutive freeze process*: with no ACK the contention window never doubles, so a station that has just transmitted may redraw a zero backoff and seize the medium one slot ahead of every deferring station, whose counter is necessarily ≥ 1. This is precisely the effect analysed by Ma and Chen [1], [2], whose closed form reproduces our measurements to within 0.36% on throughput, success probability and idle-slot occupancy; the unicast counterpart is the "anomalous slot" analysis of Tinnirello *et al.* [16], which our unicast traces independently confirm. We claim no novelty for the mechanism; our contri-

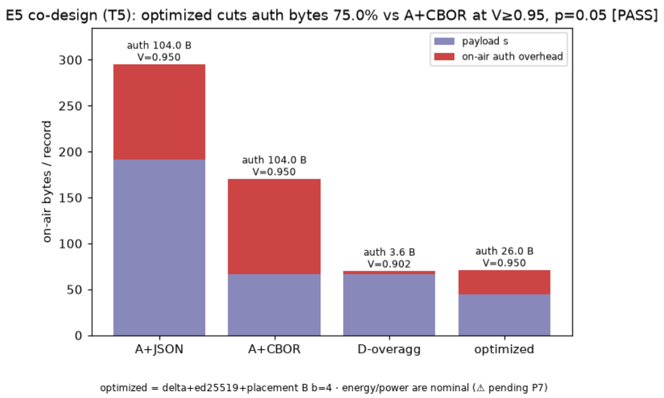

Fig. 2. E5: on-air bytes/record (payload + authentication) for the optimized configuration versus the baselines. Authentication overhead falls from 104 B to 26.0 B under the 250 ms freshness bound.

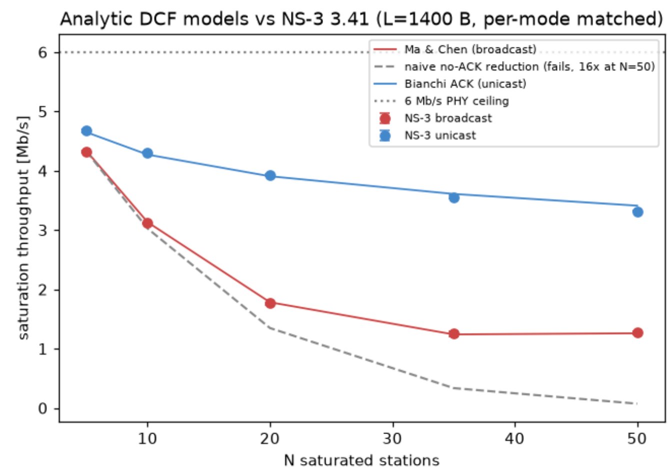

Fig. 3. Analytic DCF models vs. NS-3 3.41 saturation throughput. Unicast matches Bianchi within $+0.6/{-2.9}\%$; broadcast matches Ma and Chen [2] within 1.1%, while the naive reduction of the unicast model (dashed) fails by $16\times$ at $N{=}50$.

TABLE III
Analytic DCF models vs. NS-3 3.41, relative gap.

N	unicast vs. [14]	broadcast vs. naive reduction	broadcast vs. [2]
5	$+0.6\%$	-0.2%	-0.3%
10	$+0.9\%$	$+3.1\%$	-0.4%
20	$+0.6\%$	$+31.3\%$	-0.7%
35	-1.4%	$+273.1\%$	$+0.9\%$
50	$\mathbf{-2.9}\%$	$\mathbf{+1654}\%$	$\mathbf{+1.1}\%$

bution is its validation at $W_0{=}16$ under 802.11a timing (the original studies used $W_0{=}32, 128$ at 1 Mb/s) and a direct slot-level measurement of it.

VI. Reproducibility and Scientific Integrity

Beyond the results, the project's second product is its discipline. Anomalies were root-caused, not patched over: a stateful delta codec used statelessly (60→45 B); string-keyed CBOR replaced by canonical arrays (110→66 B); the BLS size corrected 48→96 B (with the T4 model updated); and and, on the broadcast channel, three successive explanations retracted in writing—an "$18\times$ capture" over-claim, then the capture effect itself (PHY instrumentation measured it at 0%), and finally our own claim to have discovered the responsible mechanism, which a literature check showed had been published in 2007 [1]. A whole-repo audit re-derived every formula against the code and hand-checked one point per experiment. Finally, freezing raw data risks *silent staleness*, so an automated gate re-derives every deterministic frozen artifact from the current code and fails on any drift—verified to catch the exact regression it was built for. No result is fabricated, no test skipped, no tolerance widened.

VII. Limitations and Future Work

Results are 802.11-only (a LoRa arm is future work). Energy is now measured on a Raspberry Pi 4 with INA219 meters ($P_{\text{cpu}} = 0.634$ W, $P_{\text{radio}} = 0.218$ W—both nominal values were 3–5$\times$ too high), and the byte-based headline is power-free either way. The channel models are validated only in the idealised single-collision-domain regime they describe: a spread, mobile formation introduces spatial reuse and, at long range, hidden terminals that neither model represents. We bound this: sweeping a realistic air-to-air scenario (Friis, 3D cluster, Nakagami fading) shows the idealised model is a *conservative* lower bound up to a ≈300–400 m cluster (goodput $+43$ to $+51\%$ from spatial reuse) but becomes *optimistic* by 15.7% at 500 m, where links break down; fading costs a further 5–13 points. The hardware validation is two-node. On ARM the scheme tie breaks toward Ed25519 rather than ECDSA, with an identical auth-byte cut.

VIII. Conclusion

Authentication overhead dominates tiny UAV telemetry, but it is a *co-design* problem. Jointly optimizing encoding, placement, scheme, and batching cuts on-air authentication by 75.0% at $V \geq 0.95$—meeting the falsifiable criterion—while an NS-3-validated channel model and an automated reproducibility gate keep the study honest and repeatable.

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